METHOD 18 / A COMPLETE BEGINNER GUIDE

Copulas

Question this method answers: How do two FX return series link in the *joint distribution*, separate from each leg's own volatility shape?

A copula is a model for dependence among uniform ranks u and v. It is not one Pearson-style correlation number — though copula parameters are sometimes loosely called "correlation."

Page map

Pages	Topic
2	Idea: margins vs dependence
3	FX return math (majors)
4	Ranks u, v from returns
5	Research notes (Patton 2006)
6	Symbol dictionary
7-9	Sklar, Gaussian copula, tail ideas
10-11	Teaching ranks and workflow
12	Estimation outline
13-15	Limits, practice, sources

No fitted copula parameters from live major-pair panels appear here.

Prepared for: 18 - Copulas. README status: not started.

02 / THE BASIC IDEA

Margins and copula

Each return series has its own distribution (fat tails, volatility clustering). Copula methods split the problem: model each margin, then model how uniform transforms u and v move together.

Not a single r score

Pearson r	One linear co-movement number on raw or transformed data.
Copula C	Whole dependence structure on $[0,1]^2$ — may allow asymmetric tails.
Patton (2006) FX focus	Exchange rates often show asymmetric dependence — copulas target that.

03 / RETURNS FIRST

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

1.1000 → 1.1110 ≈ +1% simple. Copula work uses many days of returns, converts each leg to pseudo-observations u_i = F■_X(x_i), v_i = F■_Y(y_i) with estimated margins F■.

Missings drop paired rows. Crisis periods change both margins and copula — report the window honestly.

04 / PSEUDO-OBSERVATIONS

From returns to u and v

```
u_i = F_hat_X(x_i)
v_i = F_hat_Y(y_i)
```

F_hat can be empirical ranks / (n+1), a fitted t-distribution CDF, or a GARCH-filtered residual CDF. Different F_hat → different u,v → different copula fit.

Section 18 in forex-formulas-explained.pdf documents the repo placement of copula dependence.

05 / PATTON (2006)

Research notes

Modelling asymmetric exchange rate dependence. International Economic Review, 47(2), 527–556. DOI: 10.1111/j.1468-2354.2006.00387.x.

Shows symmetric Gaussian copulas can miss FX dependence patterns — especially stronger co-movement during joint depreciations vs appreciations. Motivates copula families with tail asymmetry.

This guide explains concepts; parameter estimates belong in future experiment outputs with frozen code versions.

Symbol dictionary

Symbol	Meaning
F_X, F_Y	Marginal CDFs of X and Y returns.
u, v	Uniform transforms in (0,1).
C(u,v)	Copula — joint CDF on the unit square.
c(u,v)	Copula density if differentiable.
θ (theta)	Copula parameter (meaning depends on family).
Sklar	Theorem linking joint CDF to margins + copula.

07 / JOINT CDF FACTORIZATION

Sklar's theorem

```
F_XY(x,y) = C( F_X(x), F_Y(y) )
```

Any bivariate CDF with continuous margins can be written this way. Dependence lives entirely in C; margins describe each leg alone.

For discrete binned returns, copula theory needs adjustments — FX daily work usually stays with continuous or pseudo-continuous u,v.

Gaussian copula (symmetric baseline)

```
C_Gauss(u,v; rho) = Phi_2( Phi^{-1}(u), Phi^{-1}(v); rho )
```

Phi is a standard normal CDF. rho plays a role like linear correlation in the Gaussian world, but tail behaviour is symmetric — often a poor FX tail match per Patton (2006).

Use Gaussian copula as a teaching baseline, not as the final word on majors.

09 / WHY FX CARES

Tail dependence intuition

Lower-tail dependence: do both returns crash together more often than Gaussian predicts? Upper-tail dependence: joint spikes? Asymmetric copulas (clayton, rotated, or time-varying C) encode different tail stories.

One Pearson r cannot separate margin fat tails from tail dependence — copula + margin split is the point.

Teaching ranks

Day	x (return)	y (return)	u rank	v rank
1	-0.02	-0.015	0.167	0.167
2	-0.01	-0.005	0.333	0.333
3	0.0	0.005	0.500	0.500
4	0.01	0.012	0.667	0.667
5	0.02	0.018	0.833	0.833

Ranks use i/(n+1) on sorted values — a simple empirical uniform for teaching. Monotone co-movement pushes u and v up together across days.

11 / REPO USAGE

Daily majors workflow

1. Filter margins first

GARCH or empirical CDF — pick one pipeline and version it in experiments.

2. Choose copula family before seeing pair rankings

Document symmetric vs asymmetric candidates and selection criteria (AIC, tail tests).

3. Report dependence separately from volatility

A day with huge |return| affects margins; copula parameters summarize linkage among u,v after margins absorb scale.

Store outputs in experiments/v001/ when built — not in this PDF.

Estimation outline

```
Step 1: Fit F_hat_X, F_hat_Y on return samples
Step 2: Set u_i, v_i and maximize copula log-likelihood for theta
```

Time-varying copulas (Patton and follow-ups) let theta_t evolve — advanced topic for later experiment notes.

Libraries: copulas (Python), rugarch + copula packages in R — verify Sklar conventions in docs.

Limits

A copula fit is only as good as margins and the chosen family. Small n makes tail parameters unstable. Regime shifts break constant-copula models.

Copula dependence is not TE directionality and not a hedge ratio. Trading claims need out-of-sample checks outside this guide.

When margins are mis-specified, u and v can look uniform while still hiding bad tail fit — always check margin residuals before trusting theta.

Practice

Q: Is copula rho the same as Pearson r? **A:** Not in general — related under Gaussian specials, different objects otherwise.

Q: Can u and v come from price levels? **A:** Usually model return margins, not trending prices.

Q: Does asymmetric copula imply causation? **A:** No — it describes joint tail linkage in the window.

Check: middle-row teaching u ≈ 0.500, v ≈ 0.500 — both near 0.5 for balanced ranks.

Sources

[1] Patton, A. J. (2006). Modelling asymmetric exchange rate dependence. International Economic Review, 47(2), 527–556.
FX asymmetric dependence motivation.
Open source

[2] Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. Publications de l'Institut de Statistique de l'Université de Paris, 8, 229–231.
Sklar theorem original.
Open source

[3] forex-formulas-explained.pdf — section 18 (this repository).
Project formula sheet.
Open source

Rebuild: python build_guide.py. End of copulas beginner guide.